| | $\mathcal{K}_{0^+}^{\#1}$ | $\mathcal{K}_{0^-}^{\#1}{}_{\alpha\beta\chi}$ |
|---|---|---|
| $\mathcal{K}_{0^+}^{\#1}\dagger$ | $-\frac{3k^2\,{}^{(4)}\kappa_1}{2}$ | 0 |
| $\mathcal{K}_{0^-}^{\#1}\dagger^{\alpha\beta\chi}$ | 0 | $3k^2\,{}^{(4)}\kappa_{15}+3\,{}^{(2)}\kappa_1$ |

| | $\mathcal{K}_{1^+}^{\#1}{}_{\alpha\beta}$ | $\mathcal{K}_{1^+}^{\#2}{}_{\alpha\beta}$ | $\mathcal{K}_{1^-}^{\#1}{}_{\alpha}$ | $\mathcal{K}_{1^-}^{\#2}{}_{\alpha}$ |
|---|---|---|---|---|
| $\mathcal{K}_{1^+}^{\#1}\dagger^{\alpha\beta}$ | $\Upsilon_1 k^2+2\,{}^{(2)}\kappa_1$ | $\frac{\Upsilon_1 k^2+2\,{}^{(2)}\kappa_1}{\sqrt{2}}$ | 0 | 0 |
| $\mathcal{K}_{1^+}^{\#2}\dagger^{\alpha\beta}$ | $\frac{\Upsilon_1 k^2+2\,{}^{(2)}\kappa_1}{\sqrt{2}}$ | $\frac{1}{2}(\Upsilon_1 k^2+2\,{}^{(2)}\kappa_1)$ | 0 | 0 |
| $\mathcal{K}_{1^-}^{\#1}\dagger^{\alpha}$ | 0 | 0 | 0 | 0 |
| $\mathcal{K}_{1^-}^{\#2}\dagger^{\alpha}$ | 0 | 0 | 0 | 0 |

| | $\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta}$ | $\mathcal{K}_{2^-}^{\#1}{}_{\alpha\beta\chi}$ |
|---|---|---|
| $\mathcal{K}_{2^+}^{\#1}\dagger^{\alpha\beta}$ | 0 | 0 |
| $\mathcal{K}_{2^-}^{\#1}\dagger^{\alpha\beta\chi}$ | 0 | 0 |

| | $\mathcal{J}_{0^+}^{\#1}$ | $\mathcal{J}_{0^-}^{\#1}{}_{\alpha\beta\chi}$ |
|---|---|---|
| $\mathcal{J}_{0^+}^{\#1}\dagger$ | $-\frac{2}{3k^2\,{}^{(4)}\kappa_1}$ | 0 |
| $\mathcal{J}_{0^-}^{\#1}\dagger^{\alpha\beta\chi}$ | 0 | $\frac{1}{3(k^2\,{}^{(4)}\kappa_{15}+{}^{(2)}\kappa_1)}$ |

| | $\mathcal{J}_{1^+}^{\#1}{}_{\alpha\beta}$ | $\mathcal{J}_{1^+}^{\#2}{}_{\alpha\beta}$ | $\mathcal{J}_{1^-}^{\#1}{}_{\alpha}$ | $\mathcal{J}_{1^-}^{\#2}{}_{\alpha}$ |
|---|---|---|---|---|
| $\mathcal{J}_{1^+}^{\#1}\dagger^{\alpha\beta}$ | $\frac{2}{3\,\mathrm{Det}(1^+)}$ | $\frac{\sqrt{2}}{3\,\mathrm{Det}(1^+)}$ | 0 | 0 |
| $\mathcal{J}_{1^+}^{\#2}\dagger^{\alpha\beta}$ | $\frac{\sqrt{2}}{3\,\mathrm{Det}(1^+)}$ | $\frac{1}{3\,\mathrm{Det}(1^+)}$ | 0 | 0 |
| $\mathcal{J}_{1^-}^{\#1}\dagger^{\alpha}$ | 0 | 0 | 0 | 0 |
| $\mathcal{J}_{1^-}^{\#2}\dagger^{\alpha}$ | 0 | 0 | 0 | 0 |

| | $\mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta}$ | $\mathcal{J}_{2^-}^{\#1}{}_{\alpha\beta\chi}$ |
|---|---|---|
| $\mathcal{J}_{2^+}^{\#1}\dagger^{\alpha\beta}$ | 0 | 0 |
| $\mathcal{J}_{2^-}^{\#1}\dagger^{\alpha\beta\chi}$ | 0 | 0 |

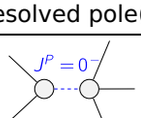
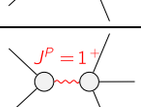
Abbreviations used in matrices

$$\Upsilon_1 == 2\,{}^{(4)}\kappa_{15} - {}^{(4)}\kappa_3 \;\&\&\; \mathrm{Det}(1^+) == k^2\left(3\,{}^{(4)}\kappa_{15} - \frac{3\,{}^{(4)}\kappa_3}{2}\right) + 3\,{}^{(2)}\kappa_1$$

Lagrangian

$$\begin{aligned}&{}^{(2)}\kappa_1\,\mathcal{K}_{\alpha\beta\chi}\,\mathcal{K}^{\alpha\beta\chi} - 2\,{}^{(2)}\kappa_1\,\mathcal{K}^{\alpha\beta\chi}\,\mathcal{K}_{\beta\alpha\chi} + {}^{(4)}\kappa_1\,\partial_\beta\mathcal{K}^{\delta}{}_{\chi\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + {}^{(4)}\kappa_3\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\alpha\chi}{}^{\delta} - 2\,{}^{(4)}\kappa_3\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta} - \\ &{}^{(4)}\kappa_3\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} - \frac{1}{2}\,{}^{(4)}\kappa_3\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi} + {}^{(4)}\kappa_{15}\,\partial_\delta\mathcal{K}_{\alpha\beta\chi}\,\partial^\delta\mathcal{K}^{\alpha\beta\chi} - 2\,{}^{(4)}\kappa_{15}\,\partial_\delta\mathcal{K}_{\beta\alpha\chi}\,\partial^\delta\mathcal{K}^{\alpha\beta\chi}\end{aligned}$$

| Added source term(s): | $\mathcal{K}^{\alpha\beta\chi}\;\mathcal{J}_{\alpha\beta\chi}$ | |
|---|---|---|
| Source constraint(s) | # constraint(s) | Covariant form |
| $\mathcal{J}_{1^-}^{\#2\,\alpha} == 0$ | 3 | $\partial_\chi\partial_\beta\mathcal{J}^{\beta\alpha\chi} == 0$ |
| $\mathcal{J}_{1^-}^{\#1\,\alpha} == 0$ | 3 | $\partial_\chi\partial^\alpha\mathcal{J}^{\beta}{}_{\beta}{}^{\chi} + \partial_\chi\partial^\chi\mathcal{J}^{\beta\alpha}{}_{\beta} == \partial_\chi\partial_\beta\mathcal{J}^{\beta\alpha\chi}$ |
| $\mathcal{J}_{1^+}^{\#1\,\alpha\beta} == \mathcal{J}_{1^+}^{\#2\,\alpha\beta}$ | 3 | $3\,\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\beta\alpha\chi} + 2\,\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\chi\alpha\beta} == 3\,\partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\alpha\beta\chi}$ |
| $\mathcal{J}_{2^-}^{\#1\,\alpha\beta\chi} == 0$ | 5 | $3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta}{}_{\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\alpha\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\beta\epsilon} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\alpha}{}_{\delta} == 3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha}{}_{\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\beta\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\alpha\epsilon} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\beta}{}_{\delta}$ |
| $\mathcal{J}_{2^+}^{\#1\,\alpha\beta} == 0$ | 5 | $3\,\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta} + 3\,\partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta} + 2\,\eta^{\alpha\beta}\,\partial_\epsilon\partial^\epsilon\partial_\delta\mathcal{J}^{\chi}{}_{\chi}{}^{\delta} == 2\,\partial_\delta\partial^\beta\partial^\alpha\mathcal{J}^{\chi}{}_{\chi}{}^{\delta} + 3\,(\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\alpha\beta\chi} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\beta\alpha\chi})$ |
| Total # constraint(s): | 19 | |

| Resolved pole(s) | # polarization(s) | Square mass | Residue |
|---|---|---|---|
|   | 1 | $-\frac{{}^{(2)}\kappa_1}{{}^{(4)}\kappa_{15}}$ | $-\frac{1}{3\,{}^{(4)}\kappa_{15}}$ |
|   | 3 | $\frac{2\,{}^{(2)}\kappa_1}{-2\,{}^{(4)}\kappa_{15}+{}^{(4)}\kappa_3}$ | $\frac{2}{6\,{}^{(4)}\kappa_{15}-3\,{}^{(4)}\kappa_3}$ |
| Resolved unitarity condition(s): | | (Demonstrably impossible) | |